

FLEET RULE OF THE ROAD TEST FOR APRIL 2015

This test is based on: The International Regulations for Preventing Collisions at Sea (1972) with amendments in force up to December 2009 as detailed in "A Seaman's Guide Pocket Book of The International Regulations for Preventing Collisions at Sea" (BR 453 Supp), ATP1(E)Vol I and the Mariners Handbook (NP100).

BR454 "A Guide to the Collision Avoidance Rules" 6th Edition by Cockcroft and Lameijer was categorised as obsolescent by NCHQ in August 2011. Care should be taken when utilising the 6th Edition as the information contained within may not reflect the most recent changes to the regulations.

The test is designed to be given to any Warfare Officer, Deck officer, Officer undertaking bridge watch-keeping duties or Young Officer/Cadet. The test should be administered under written exam conditions, the results should be recorded in the ROR/training logbook for inspection by FLEET/FOST navigation staff.

In this test, any change in the status of the ship or of the environment will be indicated in the script. .

Rules 1 to 19, 34, 35 and Annex IV must be known instinctively for instant application. A thorough knowledge of the other rules is required. The pass mark is 90% (65 Marks).

Answers are provided for uniformity of marking. This test is completely scenario based.

1. You are the OOW in HMS DARING (length 152.4 meters) about to sail from Portsmouth for Cherbourg. It is daylight with fair winds and good visibility, but a cold front is forecast to pass through the area later that day.

As you slip from the jetty and commence your passage outbound you notice RFA ORANGE LEAF alongside Gosport Jetty displaying a red flag, what does this signify? (ICS) (1)

2. On passing the Gap, the XO walks in from the Bridge wing and enquires why you are not flying CBD shapes, in accordance with QHM's direction. What is the definition of a vessel CBD, and does QHM have authority to prescribe you fly such shapes? (3)

3. As you proceed outbound you note that the nav track does not proceed along the centre of the buoyed channel. How should a vessel proceed along the course of a narrow channel or fairway? (R. 9a) (2)

4. On approaching Round Tower the CO is concerned about obscured craft and asks the Navigator to sound an appropriate sound signal. What is this and how should you navigate? (R. 9f + 34e) (2)

5. The lookout reports numerous small pleasure crafts and at least one fishing boat to the east of Spit Sand Fort. Why would you expect them to keep clear of you? (R. 9b+c) (3)

6. What shall any vessel, if the circumstance of the case admit, avoid doing in a Narrow Channel? (R. 9g) (1)

7. You proceed outbound to the east towards Nab tower and you note right ahead of you a container ship on a reciprocal course. Who should give way? (R. 14a) (1)

8. What do the Rules state about if you have any doubt as to this situation? (R. 14c) (1)

9. How large an alteration should be made to avoid collision? (R. 8b) (1)

10. What do the Rules state should be avoided when altering course and/or speed to avoid collision? (R. 8b) (1)

11. Rule 8 is 'Action to Avoid Collision'. Part 'C' states that if there is sufficient sea room, alteration of course alone maybe the most effective action to avoid a close quarters situation provided that... What? (R. 8c) (3)

12. The CO asks if the container ship is constrained by her draught. How should such a vessel indicate her condition? (R. 28) (1)

13. You pass the Nab Tower and take the watch, following the nav track to the south east passing clear of the Dover TSS. You are not using the TSS but why can you not pass closer to it? (R. 10h) (1)

14. You notice you are slowly overhauling a small merchant vessel on a similar track to your own. When shall you be deemed to be overtaking? (R. 13b) (1)

15. Who should keep out of the way? (R. 13a) (1)

16. When and what action should the vessel directed to keep out of the way take? (R. 16) **(2)**
17. Despite the fine start the visibility rapidly decreases to less than 2 miles, what action should you take with regard to your speed? (R. 19b) **(2)**
18. In restricted visibility, if you hear the fog signal of another vessel apparently forward of your beam, what action should you take assuming you can not determine risk of collision? (R. 19e) **(2)**
19. The visibility increases and you return to your normal passage routine. You receive by Navtex a warning of a tug and tow which you are likely to pass close to in the evening. What lights or shapes would you expect to see if the tow was over 200m? (R. 24a, d + e + 23a) **(6)**
20. You arrive off Cherbourg and the Navigator takes the ship to anchor. What lights or shapes would you display? (R. 30) **(3)**
21. As you approach you note a small craft moving erratically on your planned track. As it fails to maintain a constant course or speed it is difficult to ascertain risk of collision. How could you indicate that you are unsure as to its intentions? (R. 34d) **(1)**

22. The small vessel returns to the coast and you anchor safely. As you handover the watch to your relief you discuss the poor visibility earlier and how you are confident it will not return overnight. If the visibility was to decrease and you remained at anchor, what sound signals would you make? (R. 35g) (2)

23. How would these signals be supplemented if you were aground? (R. 35h) (1)

After completing the diplomacy visit off Cherbourg, the ship weighs anchor and proceeds on coastal passage. It is the forenoon watch; the weather is fine with good visibility and moderate shipping.

24. You sight a vessel at GREEN 40, range 3 miles and you are on his port bow. The bearing is steady and you assess there is a risk of collision. What actions are required by the Rules? (R. 15) (1)

25. A vessel is sighted at RED 50, range 4 miles. The range is closing and the bearing is moving very slowly right. You are 40° on his starboard bow. What action is required by the Rules? (R. 15) (1)

26. You sight a vessel apparently stopped. The vessel is displaying three shapes - ball diamond ball - in a vertical line. What is it and how can you tell if the vessel is underway or at anchor? (R. 27b(iv), 30 a(i, ii)) (2)

27. You sight a large vessel displaying a cylinder where it can best be seen. What does this indicate and what action should you take if you approach this vessel? (R. 18d/ R. 28) (2)

28. You sight a single vessel displaying two balls in a vertical line. If the vessel is making way, what lights should it display by night? (R. 27a) **(2)**

29. You sight a number of small vessels, exhibiting a shape consisting of two cones with their apexes together in a vertical line. In addition, most are displaying a cone apex upwards. What are these vessels engaged, what does the cone apex upwards indicate and how should you pass them? (R. 26c) **(3)**

30. You sight MCMVs displaying the shapes for conducting mine countermeasures operations. What action should they take when approaching a helicopter in the dip? What is the helo's status? (ATP1E) **(2)**

31. Surprisingly, despite the fine start, you encounter a bank of fog ahead. When should you commence making fog signals and what should they be? (R. 35 / R. 35a) **(2)**

32. You hear a sound signal consisting of two prolonged blasts, followed shortly afterwards by four short blasts. What precisely do these sound signals indicate? (R. 35b/k) **(2)**

33. The lookout reports hearing a sound signal consisting of one short, one prolonged and one short sounded in succession. What precisely does this signify? (R. 35g) **(1)**

34. You detect a vessel on radar at GREEN 60, range 5 miles. Careful plotting shows that a risk of collision exists. How should you manoeuvre if there are no immediate navigational constraints? What shall be avoided? (R. 19d) **(2)**

35. The light is fading but the good news is that visibility is rapidly improving. When must navigation lights be exhibited? What restrictions apply to other lights? (R. 20b) **(2)**

36. You sight a vessel displaying two all-round lights in a vertical line, the upper being red and the lower white, in addition to sidelights and stern light. What is this? (R. 26c) **(1)**

37. You sight a vessel ahead showing two additional masthead lights on her foremast in addition to her normal navigation lights.

a. What specifically does this indicate? (R. 24a) **(1)**

b. Describe any additional light would you expect to see if you approached this vessel from astern? (R. 24a) **(2)**

38. You sight a vessel displaying three all-round lights in a vertical line, the highest and lowest red and the middle light white. In addition, you identify a single white light in the fore part of the vessel. What types of operation could this vessel be engaged in and what can you conclude if these are the only lights? (R. 27b) **(2)**

39. You sight two MCMVs engaged in mine clearance exercises. What lights should the vessels be displaying in addition to their normal navigation lights and how far off should you pass? (R. 27f) **(2)**

Total _____ /72

1. You are the OOW in HMS DARING (length 152.4 meters) about to sail from Portsmouth for Cherbourg. It is daylight with fair winds and good visibility, but a cold front is forecast to pass through the area later that day.

As you slip from the jetty and commence your passage outbound you notice RFA ORANGE LEAF alongside Gosport Jetty displaying a red flag, what does this signify? (ICS) (1)

She is taking in, discharging or carrying dangerous goods.

2. On passing the Gap, the XO walks in from the Bridge wing and enquires why you are not flying CBD shapes, in accordance with QHM's direction. What is the definition of a vessel CBD, and does QHM have authority to prescribe you fly such shapes? (R.3h, (3)

The term Vessel constrained by her draught means a power driven vessel (1) which, because of her draught in relation to the available depth and width of navigable water, is severely restricted in her ability to deviate from the course she is following (1). QHM are considered an appropriate Authority for roadsteads, harbours, rivers, lakes or inland waterways.(1)

3. As you proceed outbound you note that the nav track does not proceed along the centre of the buoyed channel. How should a vessel proceed along the course of a narrow channel or fairway? (R. 9a) (2)

A vessel proceeding along the course of a Narrow Channel or Fairway shall keep as near to the outer limit of the channel or Fairway which lies on her starboard side as is safe and practicable.

4. On approaching Round Tower the CO is concerned about obscured craft and asks the Navigator to sound an appropriate sound signal. What is this and how should you navigate? (R. 9f + 34e) (2)

A vessel nearing a bend or an area of a narrow channel or fairway where other vessels may be obscured by an intervening obstruction shall navigate with particular alertness and caution and shall sound the appropriate signal prescribed in Rule 34(e) – one prolonged blast.

5. The lookout reports numerous small pleasure crafts and at least one fishing boat to the east of Spit Sand Fort. Why would you expect them to keep clear of you? (R. 9b+c) (3)

A vessel of less than 20 metres in length or a sailing vessel shall not Impede the passage of a vessel which can safely navigate only within a Narrow Channel or Fairway.

A vessel engaged in fishing shall not impede the passage of any other vessel navigating within a narrow channel or fairway.

6. What shall any vessel, if the circumstance of the case admit, avoid doing in a Narrow Channel? (R. 9g) (1)

Any vessel shall, if the circumstances of the case admit, avoid anchoring in a Narrow Channel.

7. You proceed outbound to the east towards Nab tower and you note right ahead of you a container ship on a reciprocal course. Who should give way? (R. 14a) (1)

When two power-driven vessels are meeting on reciprocal or nearly reciprocal courses so as to involve risk of collision each shall alter her course to starboard so that each shall pass on the port side of the other.

8. What do the Rules state about if you have any doubt as to this situation? (R. 14c) (1)

When a vessel is in any doubt as to whether such a situation exists she shall assume that it does exist and act accordingly.

9. How large an alteration should be made to avoid collision? (R. 8b) (1)

It shall be large enough to be readily apparent to another vessel observing visually or by radar.

10. What do the Rules state should be avoided when altering course and/or speed to avoid collision? (R. 8b) (1)

A succession of small alterations of course and/or speed.

11. Rule 8 is 'Action to Avoid Collision'. Part 'C' states that if there is sufficient sea room, alteration of course alone maybe the most effective action to avoid a close quarters situation provided that... What? (R. 8c) (3)

It is made in good time, is substantial and does not result in another close-quarters situation.

12. The CO asks if the container ship is constrained by her draught. How should such a vessel indicate her condition? (R. 28) (1)

A vessel constrained by her draught may, in addition to the lights prescribed for power-driven vessels in Rule 23, exhibit where they can best be seen three all-round red lights in a vertical line, or a cylinder.

13. You pass the Nab Tower and take the watch, following the nav track to the south east passing clear of the Dover TSS. You are not using the TSS but why can you not pass closer to it? (R. 10h) (1)

A vessel not using a traffic separation scheme shall avoid it by as wide a margin as practicable.

14. You notice you are slowly overhauling a small merchant vessel on a similar track to your own. When shall you be deemed to be overtaking? (R. 13b) (1)

A vessel shall be deemed to be overtaking when coming up with another vessel from a direction more than 22.5 degrees abaft her beam, that is, in such a position with reference to the vessel she is overtaking, that at night she would be able to see only the sternlight of that vessel but neither of her sidelights.

15. Who should keep out of the way? (R. 13a) (1)

Notwithstanding anything contained in the Rules of Part B, Sections I and II, any vessel overtaking any other shall keep out of the way of the vessel being overtaken.

16. When and what action should the vessel directed to keep out of the way take? (R. 16) (2)

Every vessel which is directed to keep out of the way of another vessel shall, so far as possible, take early and substantial action to keep well clear.

17. Despite the fine start the visibility rapidly decreases to less than 2 miles, what action should you take with regard to your speed? (R. 19b) (2)

Every vessel shall proceed at a safe speed adapted to the prevailing circumstances and conditions of restricted visibility. A power-driven vessel shall have her engines ready for immediate manoeuvre.

18. In restricted visibility, if you hear the fog signal of another vessel apparently forward of your beam, what action should you take assuming you can not determine risk of collision? (R. 19e) (2)

Except where it has been determined that a risk of collision does not exist, every vessel which hears apparently forward of her beam the fog signal of another vessel, or which cannot avoid a close-quarters situation with another vessel forward of her beam, shall reduce her speed to the minimum at which she can maintain her course. She shall if necessary take all way off and in any even navigate with extreme caution until danger of collision is over.

19. The visibility increases and you return to your normal passage routine. You receive by Navtex a warning of a tug and tow which you are likely to pass close to in the evening. What lights or shapes would you expect to see if the tow was over 200m? (R. 24a, d + e + 23a ii) (6)

A power-driven vessel when towing shall exhibit:

- (i) three masthead lights in a vertical line. ;
- (ii) sidelights;
- (iii) a sternlight;
- (iv) a towing light in a vertical line above the sternlight;
- (v) a diamond shape where it can best be seen.

A second masthead light abaft of and higher than the forward one; except that a vessel of less than 50 metres in length shall not be obliged to exhibit such light but may do so;

A vessel or object being towed, other than those mentioned in paragraph (g) of this Rule, shall exhibit:

- (i) sidelights;
- (ii) a sternlight;
- (iii) when the length of tow exceeds 200 metres, a diamond shape where it can best be seen.

20. You arrive off Cherbourg and the Navigator takes the ship to anchor. What lights or shapes would you display? (R. 30) (3)

A vessel at anchor shall exhibit where it can best be seen:

- (i) in the fore part, an all-round white light or one ball;
- (ii) at or near the stern and at a lower level than the light prescribed in sub-paragraph (i), an all-round white light."

A vessel at anchor may, and a vessel of 100 metres and more in length shall, also use the available working or equivalent lights to illuminate her decks.

21. As you approach you note a small craft moving erratically on your planned track. As it fails to maintain a constant course or speed it is difficult to ascertain risk of collision. How could you indicate that you are unsure as to its intentions? (R. 34d) (1)

When vessels in sight of one another are approaching each other and from any cause either vessel fails to understand the intentions or actions of the other, or is in any doubt whether sufficient action is being taken by the other to avoid collision, the vessel in doubt shall immediately indicate such doubt by giving at least five short and rapid blasts on the whistle. Such signal may be supplemented by a light signal of at least five short and rapid flashes. (R. 34d)

22. The small vessel returns to the coast and you anchor safely. As you handover the watch to your relief you discuss the poor visibility earlier and how you are confident it will not return overnight. If the visibility was to decrease and you remained at anchor, what sound signals would you make? (R. 35g) (2)

A vessel at anchor shall

- at intervals of not more than one minute

- ring the bell rapidly for about 5 seconds in the forepart of the vessel and, immediately after ringing of the bell, the gong shall be sounded rapidly for about 5 seconds in the after part of the vessel.

23. How would these signals be supplemented if you were aground? (R. 35h) (1)

A vessel aground shall give the bell signal and if required the gong signal prescribed in paragraph (g) of this Rule and shall, in addition, give three separate and distinct strokes on the bell immediately before and after the rapid ringing of the bell. A vessel aground may in addition sound an appropriate whistle signal.

After completing the diplomacy visit off Cherbourg, the ship weighs anchor and proceeds on coastal passage. It is the forenoon watch; the weather is fine with good visibility and moderate shipping.

24. You sight a vessel at GREEN 40, range 3 miles and you are on his port bow. The bearing is steady and you assess there is a risk of collision. What actions are required by the Rules? (R. 15) (1)

...the vessel which has the other on her starboard side shall keep out of the way. Sound one short blast and alter course to starboard.

25. A vessel is sighted at RED 50, range 4 miles. The range is closing and the bearing is moving very slowly right. You are 40° on his starboard bow. What action is required by the Rules? (R. 15) (1)

Stand on but monitor the give-way vessel:

(When two power-driven vessels are crossing so as to involve a risk of collision, the vessel which has the other on her own starboard side shall keep out of the way and shall, if the circumstances of the case admit, avoid crossing ahead of the other vessel.)

26. You sight a vessel apparently stopped. The vessel is displaying three shapes - ball diamond ball - in a vertical line. What is it and how can you tell if the vessel is underway or at anchor? (R. 27b(iv), 30 a(i, ii)) (2)

The vessel is “restricted in her ability to manoeuvre”. If at anchor, she would display in the fore part one ball. The movement can also be determined by bow wave/wake (if present) and you should keep well clear.

27. You sight a large vessel displaying a cylinder where it can best be seen. What does this indicate and what action should you take if you approach this vessel? (R. 18d/ R. 28) (2)

A vessel constrained by her draught may, in addition to the lights prescribed for a power-driven vessel exhibit a cylinder where it can best be seen. Rule 18d A power-driven vessel underway shall if the circumstances of the case admit, avoid impeding the safe passage of a vessel constrained by her draught, exhibiting the signals in Rule 28.

28. You sight a single vessel displaying two balls in a vertical line. If the vessel is making way, what lights should it display by night? (R. 27a) (2)

The vessel is “not under command”. At night she should display two all-round red lights (in a vertical line) plus sidelights and sternlight. (Masthead lights should be extinguished.)

29. You sight a number of small vessels, exhibiting a shape consisting of two cones with their apexes together in a vertical line. In addition, most are displaying a cone apex upwards. What are these vessels engaged, what does the cone apex upwards indicate and how should you pass them? (R. 26c) (3)

A vessel when engaged in fishing, other than trawling, shall exhibit a shape consisting of two cones with apexes together in a vertical line one above the other. When there is outlying gear extending more than 150m horizontally from the vessel, a cone apex upwards in the direction of the gear. Pass more than 150 m on the side of the outlying gear.

30. You sight MCMVs displaying the shapes for conducting mine countermeasures operations. What action should they take when approaching a helicopter in the dip? What is the helo’s status? (ATP1E) (2)

A helicopter in the dip is considered as a ship not under command and no vessels should approach within 500 yards.

31. Surprisingly, despite the fine start, you encounter a bank of fog ahead. When should you commence making fog signals and what should they be? (R. 35 / R. 35a) (2)

As you approach the bank of fog. Rule 35. In or near an area of restricted visibility, whether by day or night, the signals prescribed in this Rule shall be used. Rule 35a. A power-driven vessel making way through the water shall sound at intervals of not more than 2 minutes one prolonged blast.

32. You hear a sound signal consisting of two prolonged blasts, followed shortly afterwards by four short blasts. What precisely do these sound signals indicate? (R. 35b/k) (2)

A power-driven vessel underway but not making way which also sounds an identity signal indicating a pilot vessel when engaged on pilotage duty.

33. The lookout reports hearing a sound signal consisting of one short, one prolonged and one short sounded in succession. What precisely does this signify? (R. 35g) (1)

A vessel at anchor may be in addition sound three blasts in succession, namely one short, one prolonged and one short blast, to give warning of her position and of the possibility of collision to an approaching vessel.

34. You detect a vessel on radar at GREEN 60, range 5 miles. Careful plotting shows that a risk of collision exists. How should you manoeuvre if there are no immediate navigational constraints? What shall be avoided? (R. 19d) (2)

She shall take avoiding action in ample time, provided that when such action consists of an alteration of course, so far as possible, an alteration of course to port for a vessel forward of the beam other than for a vessel being overtaken shall be avoided.

35. The light is fading but the good news is that visibility is rapidly improving. When must navigation lights be exhibited? What restrictions apply to other lights? (R. 20b) (2)

The Rules concerning lights must be complied with from sunset to sunrise, and during such time no other lights shall be exhibited, except such lights as cannot be mistaken for the lights specified in these Rules or do not impair their visibility or distinctive character, or interfere with the keeping of a proper look-out.

36. You sight a vessel displaying two all-round lights in a vertical line, the upper being red and the lower white, in addition to sidelights and stern light. What is this? (R. 26c) (1)

A vessel engaged in fishing, other than trawling, when making way through the water.

37. You sight a vessel ahead showing two additional masthead lights on her foremast in addition to her normal navigation lights.

a. What specifically does this indicate? (R. 24a) (1)

A power-driven vessel when towing; the length of tow, measured from the stern of the towing vessel to the after end exceeds 200 metres.

b. Describe any additional light would you expect to see if you approached this vessel from astern? (R. 24a) (2)

A power-driven vessel when towing shall exhibit: a towing light in a vertical line above the sternlight (yellow light having the same characteristics as the sternlight). In addition, a vessel or object being towed shall exhibit sidelights and a sternlight (or lights for an inconspicuous partly submerged vessel or object).

38. You sight a vessel displaying three all-round lights in a vertical line, the highest and lowest red and the middle light white. In addition, you identify a single white light in the fore part of the vessel. What types of operation could this vessel be engaged in and what can you conclude if these are the only lights? (R. 27b) (2)

A vessel restricted in her ability to manoeuvre, except a vessel engaged in mine-clearance operations, shall exhibit three all-round lights in a vertical line where they can best be seen. The highest and lowest of these lights shall be red and the middle light shall be white. When at anchor, in addition, the light(s) prescribed in Rule 30. Could be laying, servicing or picking up a navigation mark, submarine cable or pipeline, or surveying but unlikely to be dredging or carrying out underwater operations due to the anchor light.

39. You sight two MCMVs engaged in mine clearance exercises. What lights should the vessels be displaying in addition to their normal navigation lights and how far off should you pass? (R. 27f) (2)

A vessel engaged in mine clearance operations shall in addition to the lights prescribed for a power-driven vessel in Rule 23 or to the lights or shapes prescribed for a vessel at anchor

in Rule 30 as appropriate, exhibit three all-round green lights. One of these lights shall be exhibited near the foremast head and one at each end of the fore yard.
You should pass at least 1000m away from a vessel engaged in mine-clearance operations.

Total _____ /72